Mathematical Foundations of Political Methodology

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Confidence intervals

- Interval Estimation
- Hypothesis Testing
- Power and Confidence
- P values

Interval Estimation

- An interval estimator maps a random variable onto an interval, rather than a point.
- If our estimator is $\hat{\theta}$, an interval estimator is written

$$[\hat{\theta}_{lower},\ \hat{\theta}_{upper}]$$

- The interval represents the range of possible values within which we are confident to find the true θ.
- This is often called a confidence interval.

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Interval Estimation

Definition

A confidence interval with a confidence level $1-\alpha$ *is an interval estimator* $[\hat{\theta}_{lower},\ \hat{\theta}_{upper}]$*, that is guaranteed to cover the true parameter* $100 \times (1-\alpha)$ *of the time.*

- Coverage probability describes how likely it is that the interval covers the true parameter value.
- Shorter intervals are better.

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Normal Population with known σ^2

- Suppose we have n draws from a Normal distribution $N(\mu,\ 2)$

$$\frac{\bar{X}_n - \mu}{2/\sqrt{n}} \sim N(0,1)$$

- Using the inference from before,

$$F(1.96) = P(\frac{|\bar{X}_n - \mu|}{2/\sqrt{n}} < 1.96) = .95$$

$$P(-1.96 < \frac{\bar{X}_n - \mu}{2/\sqrt{n}} < 1.96) = .95$$

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- A confidence interval is just a rearrangement:

$$P(-1.96 < \frac{\bar{X}_n - \mu}{2/\sqrt{n}} < 1.96) = .95$$

$$P(-1.96 * 2/\sqrt{n} < \bar{X}_n - \mu < 1.96 * 2/\sqrt{n}) = .95$$

$$P(\bar{X}_n - 1.96 * 2/\sqrt{n} < \mu < \bar{X}_n + 1.96 * 2/\sqrt{n}) = .95$$

- So the following interval estimator contains the true population μ with probability .95:

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Simulations

```
> library(animation)
> conf.int()
```

Example Normal Population with known σ^2

- Suppose that the WHO is enacting a new Ebola monitoring service at 400 airports which may hold up lines.
- Suppose that we know that the ebola monitoring service has historically had a variance of 9 people per hour and is normally distributed.
- We surveyed 40 of these monitors, and found that the sample had an average speed of 70 people per hour.
- What is the estimate of the speed of the service?
- The following interval estimator contains the true population μ with probability .95:

$$[\bar{X}_n - 1.96 * \sigma/\sqrt{n}, \bar{X}_n + 1.96 * \sigma/\sqrt{n}]$$

- The following interval estimate has a confidence level of .95:

$$[70 - 1.96 * 3/\sqrt{40}, 70 + 1.96 * 3/\sqrt{40}]$$

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The shortest interval

- If $\hat{\theta} \sim N(\mu, \sigma)$

$$[\hat{\theta} - z_{\alpha/2} * \sqrt{var(\hat{\theta})},\ \hat{\theta} + z_{\alpha/2} * \sqrt{var(\hat{\theta})}]$$

- If $\hat{\theta} \sim t_{n-1}$

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Example Normal Population with unknown σ^2

- Suppose that we are trying to find the average height of a village in Afghanistan.
- Suppose that we know that altitude is normally distributed.
- We surveyed 20 villages, and found that the sample had an average altitude of 3200 meters with a sample variance of 1200 meters.
- What is the estimate of the average altitude in Afghanistan?
- We use the t distribution with 19 degrees of freedom.
- The following interval estimate has a confidence level of .95:

$$[3200 - t_{19}(\alpha/2) * \sqrt{1200)}/\sqrt{20},\ 3200 + t_{19}(\alpha/2) * \sqrt{1200}/\sqrt{20}]$$

$$[3200 - 2.09 * \sqrt{1200)}/\sqrt{20},\ 3200 + 2.09 * \sqrt{1200}/\sqrt{20}]$$

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Interpretation

- A 95% confidence interval is an estimate of plausible values for the population parameter.
- It is not a measure of the degree of probability, belief, or support that an unknown parameter value lies in *this* specific interval.
- A particular confidence interval of 95% does not mean that there is a 95% probability of a sample mean from a repeat of the experiment falling within *this* interval.
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Confidence interval for the difference in means with known variances

- Suppose that we are trying to find the average height of a village in Afghanistan relative to a village in Pakistan.
- Suppose that we know that altitude is normally distributed, and the variance is 1200 meters and 200 meters respectively.
- We surveyed 20 villages in Afghanistan and 30 villages in Pakistan, $\bar{X}_A = 3200, \bar{X}_P = 1200$
- What is the estimate of the difference in average altitudes?
- The difference in Normals is normal.

$$\bar{X}_A - \bar{X}_P \sim N\left(\mu_A - \mu_P, \sqrt{(\frac{\sigma_A^2}{n_A} + \frac{\sigma_P^2}{n_P})}\right)$$

- The following interval estimate has a confidence level of .95:

$$[\bar{X}_A - \bar{X}_P - z_{\alpha/2} * \sqrt{(\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y})},\ \bar{X}_A - \bar{X}_P + z_{\alpha/2} * \sqrt{(\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y})}]$$

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Hypothesis Tests

- Point estimation: what is the most likely value of the parameter?
- Interval Estimation: what is the range of plausible values of the parameter?
- Hypothesis testing: do we have evidence to reject a hypothesized interval of parameter values?
- Failure to find such evidence does not suggest that the hypothesis is true.

Definition of Hypothesis

Definition

A *hypothesis* *is a statement about a population parameter*

- Does democracy promotion promote peace? (lower rates of conflict among treatment group).
- Do treaties constrain countries? (is behavior distinct among treaty signers?)
- Is support for Trump above 50%?
- Is Ohio State good this year?

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A *Hypothesis Test* *compares two mutually exclusive sets of parameters,*

- *Null hypothesis:* H_0
- *Alternative hypothesis:* H_1

- Does democracy promotion promote peace?
  - H_0: Democracy promotion does not promote peace (treated and untreated groups are equally conflictual).
  - H_1: Democracy promotion promotes peace (treated and untreated groups are different).
- Is Ohio State good this year?
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Testing Procedure

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Given a test statistic $t(X)$, a hypothesis testing procedure is a rule that specifies

- *Value of test statistic when we do not reject H_0: acceptance region*
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- *Value separating regions: critical value*

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- I had a theory that students should study for 10 hours a week.
- Is that theory consistent with practice?
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- Failing to reject the null does not mean that the null is true.

Example: Homework

- We can estimate $S^2 = \frac{1}{N-1}\sum_{i=1}^{N}(X_i - \bar{X})^2$

$$\frac{\bar{X} - \mu}{S/\sqrt{N}} \sim t_{N-1}$$

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Error Types

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- Type I Error: reject a null that is true

  Deciding the homework is too hard when it is not.
- Type II Error: failing to reject a null that is false.

  Leaving the homework the same when it is too hard.

The Size of test (α) is the probability of Type I Error (false discovery).
The Power of test is 1 - the probability of Type II Error (β) (failed discovery).
What is worse: False discovery or missed detection?
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Example of Test Statistic and Critical Region

- Set $\mu = 10$
- $t(X) \equiv \frac{\bar{X}-10}{s/\sqrt{N}} \sim t_{N-1}$
- Critical value (x_0): how long would you have to take on average for me to feel that you are working too hard?
- $P(|t(X)| > x_0) = \alpha$
- $P(t(X) < -x_0) + P(t(X) > x_0) = \alpha$
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- Set $\mu = 10$ for 22 students.
- Set $\alpha = .05$
- Using the t distribution, x_0 is 2.07
- Suppose the sample mean is 12 hours.
- Suppose the sample standard deviation is 2.6 hours.

$$P(t(X) > x_0)$$

$$\frac{12 - 10}{2.6/\sqrt{22}} > 2.07$$

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$$\text{p-value} = \int_{t(X)}^{\infty} f(x)dx = 0.0008$$

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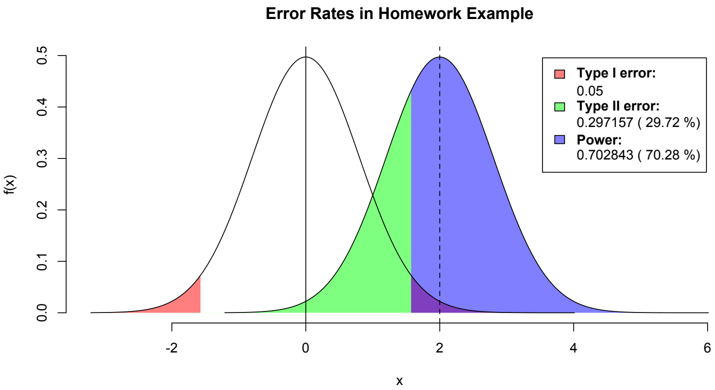

Error Rates in Homework Example
f(x)
0.0
0.1
0.2
0.3
0.4
0.5
-2
0
2
4
6
x
Type I error:
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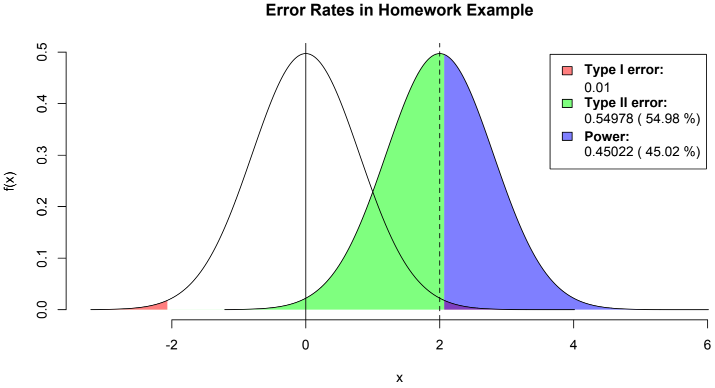
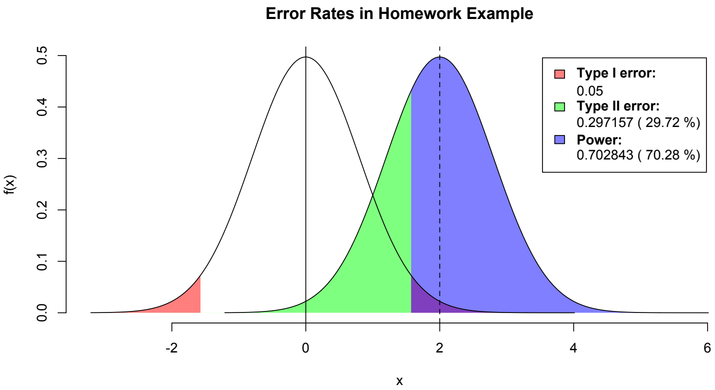
Type II error:
0.297157 (29.72 %)
Power:
0.702843 (70.28 %)

Increase α





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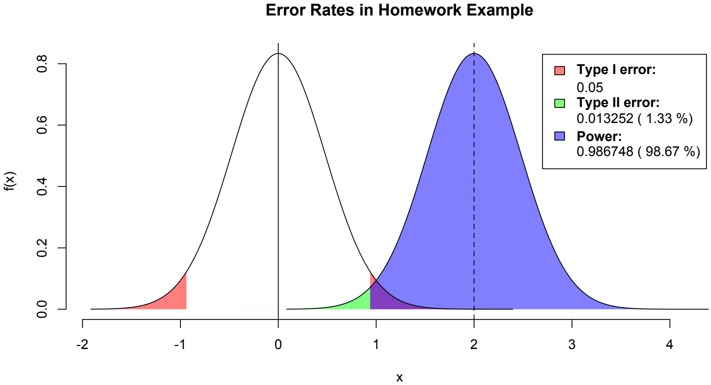

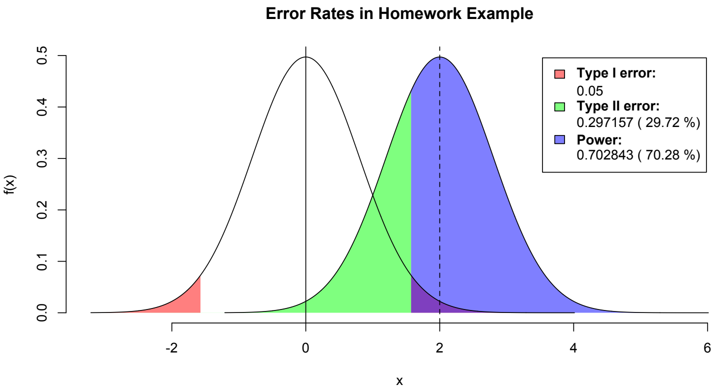
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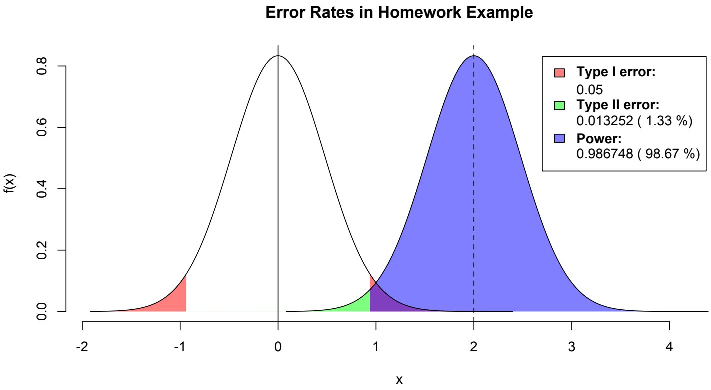

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Power will depend on four factors:

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Confidence Intervals and Hypothesis testing.

- If $\hat{\theta} \sim N(\mu, \sigma)$, the confidence intervals are:

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If $X_1,\ X_2,\ X_3, \ldots X_n$ is a random sample from a $N(\mu\sigma^2)$ data, $\frac{\bar{X}-\mu_0}{S/\sqrt{n}}$ has a students t distribution with $n-1$ degrees of freedom.

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What to do?

- Report more information about the distribution.
- Design studies so that you care about the estimates, not just rejecting a null hypothesis.